Bleeding/Bleeding Diathesis

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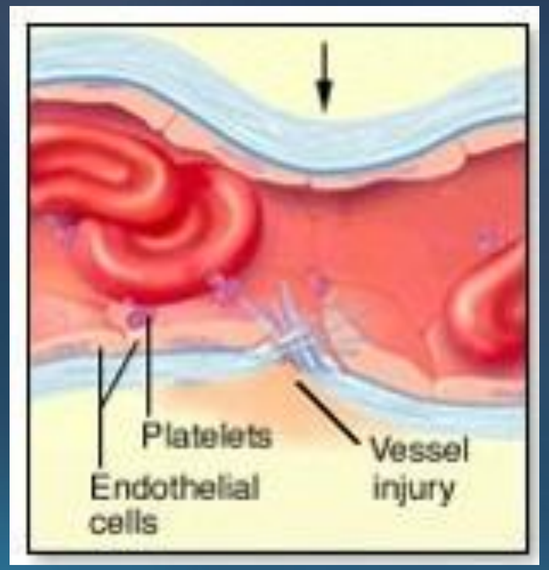
Objectives

1. Hemostasis
2. Hemostatic Agents
3. Postoperative Complication

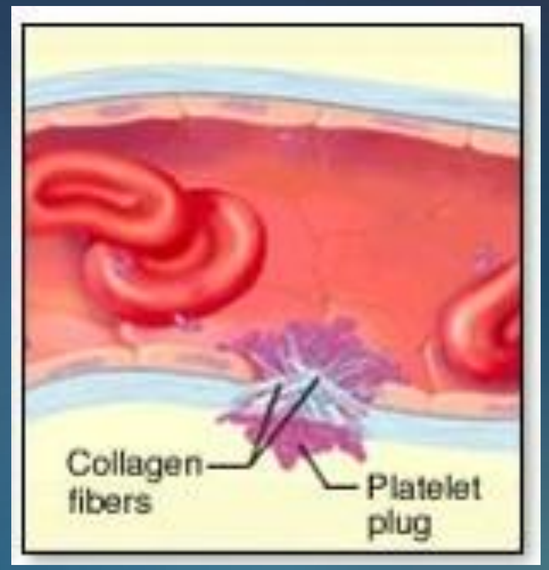
Hemostasis

1. Vascular
2. Platelet (Primary Hemostasis)
3. Coagulation Cascade (Secondary Hemostasis)
4. Fibrinolytic

Mechanisms of hemostasis



Vasoconstriction



Platelet activation

Fibrin production

Mechanisms of hemostasis

Vasoconstriction

- Vessel smooth muscle contraction mediated by serotonin and other platelet and endothelium locally released mediators aiming at reducing the blood flow and minimizing the blood loss
- Endothelins released from endothelial cells

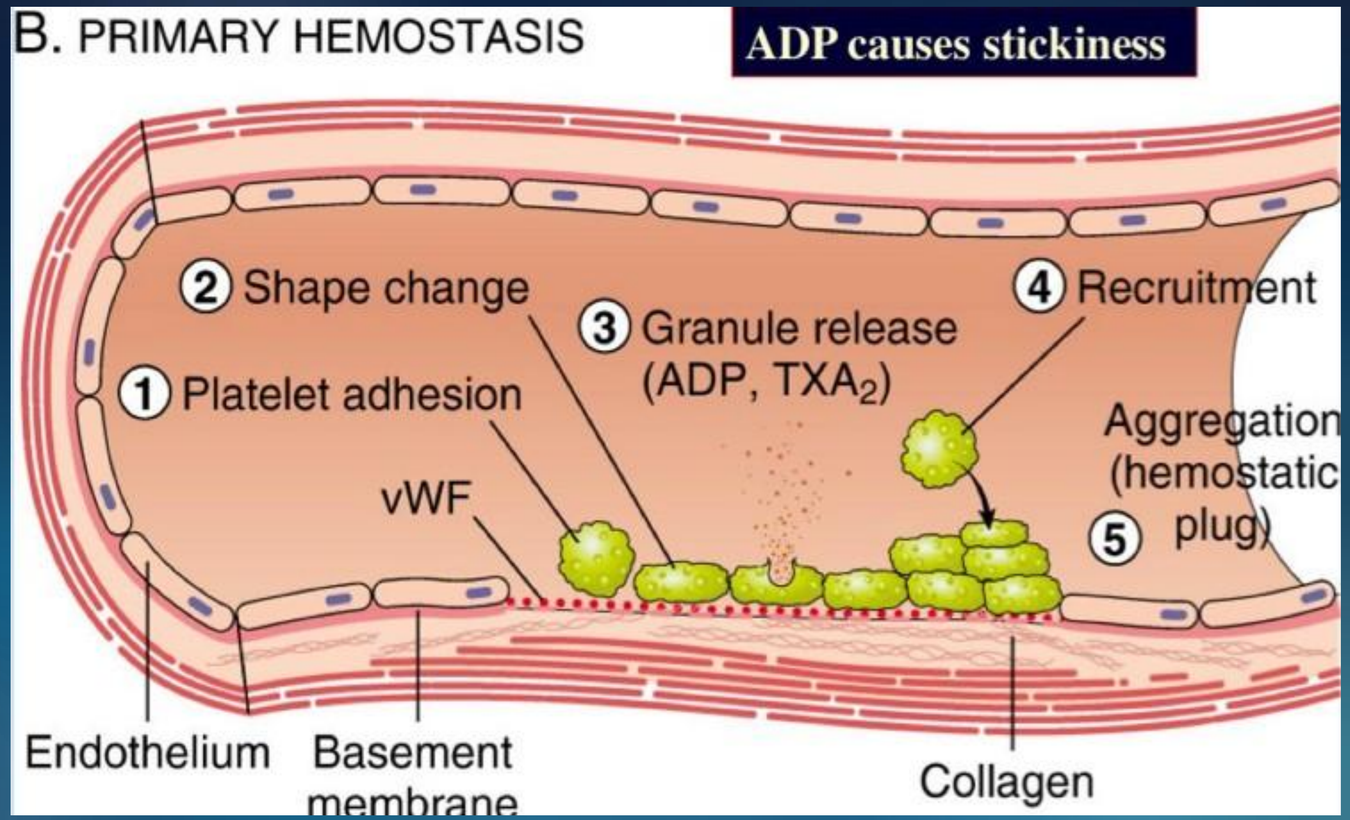
Platelet activation

- Induced by Von Willebrand factor (endothelial cells), exposed sub endothelial collagen, thrombin.
- Causes platelet aggregation and release of all kinds of locally acting mediators
- Leads to the formation of the platelet plug (primary hemostasis)

Fibrin production

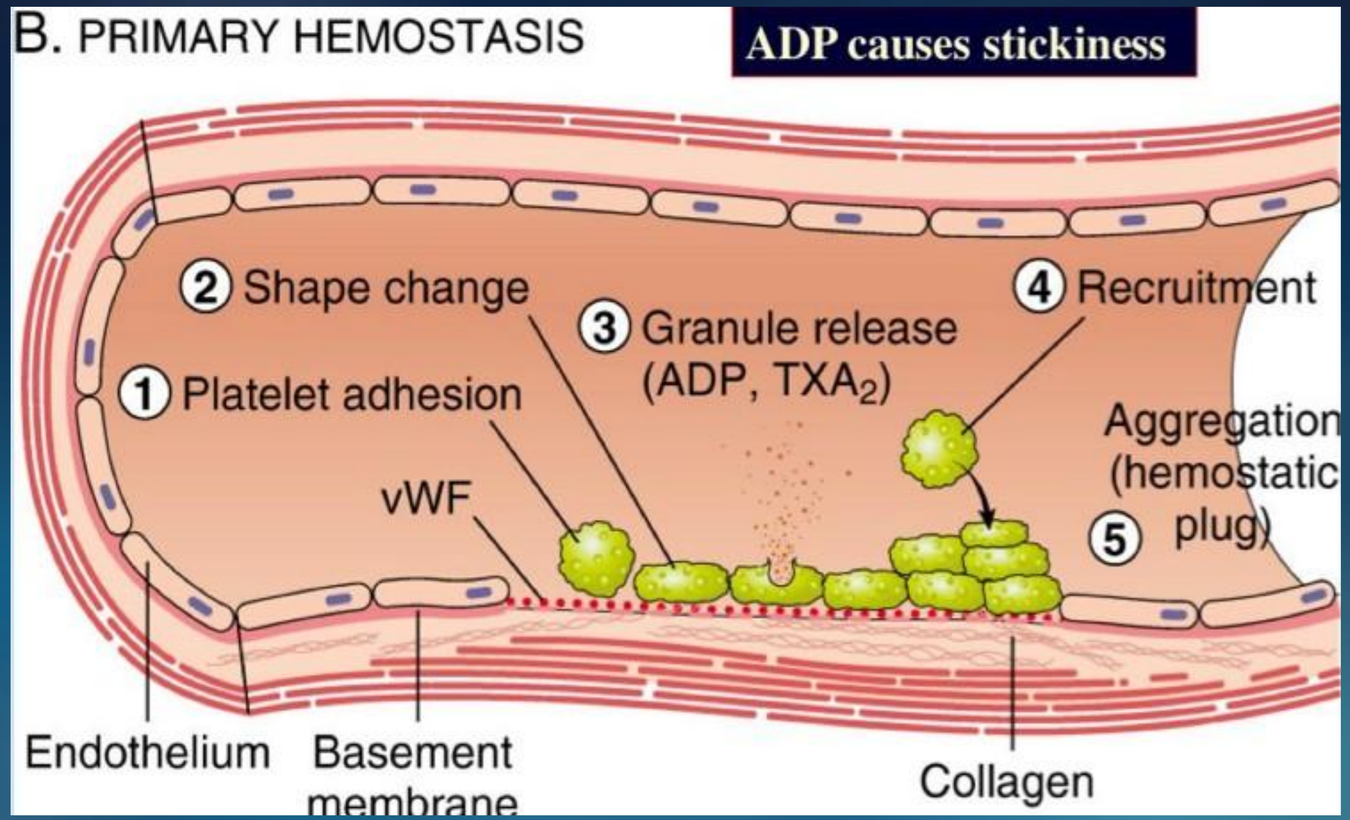
- Initiated by activation of coagulation cascade factors by local factors such as sub-endothelial collagen and tissue proteins.
- Leads to the formation of the hemostatic plug (secondary hemostasis)

Mechanisms of hemostasis



- Damaged vessels expose subendothelial collagen which binds VWF
- Platelets bind to VWF on damaged endothelium through receptor GP1b and become "activated"
- Activated platelets release mediators such as ADP and TXA2, activating more platelets and allowing GPIIb/GPIIIa receptors on platelets to bind fibrinogen, which aggregates platelets together

Drugs and Hemostasis



- ASA – TXA2
- Clopidogrel – Plavix, Ticlodipine Ticlid, Ticagrelor all of them interfere with ADP
- Abciximab – GPIIb/IIIa

Measurement:

- Quantity Platelet count
- Quality: Bleeding time

- Note: Ginseng is an antiplatelet
- Saw Palmetto is an anticoagulant

Vessels

Vessels

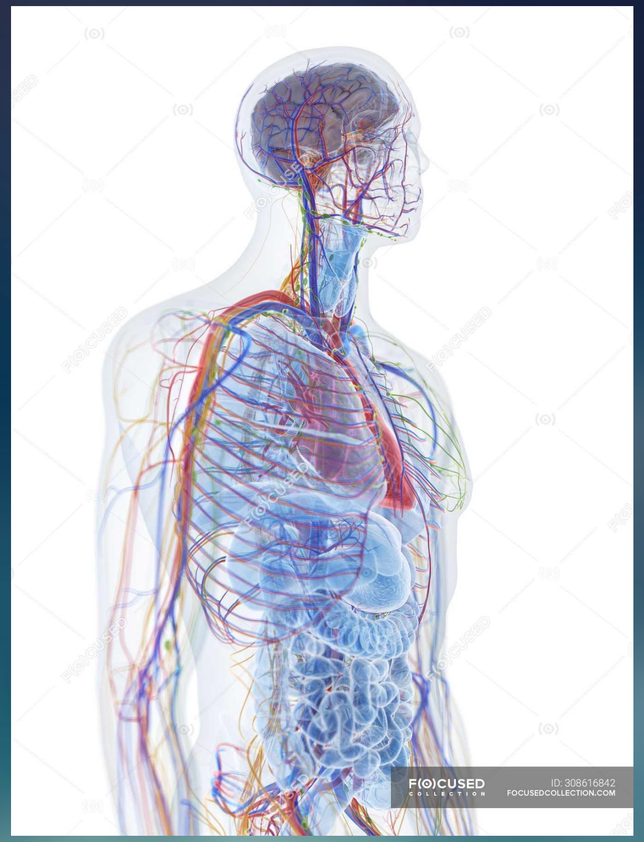
Structural abnormalities of vessel walls can give rise to bleeding diathesis

Nonthrombocytopenic purpuras (purpura: Presence of purple spots in the skin due to small vessel bleeding)

Vasculitis: Henoch-Schonlein Purpura
Microvascular injury: Senile Purpura, Hypertension

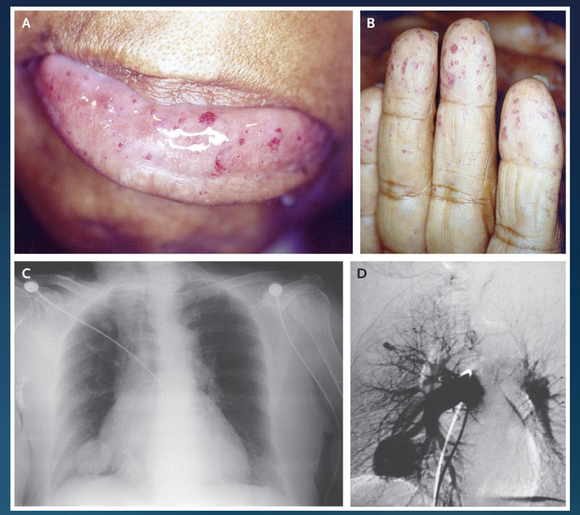
Collagen disorders
Scurvy: Vit. C deficiency affecting collagen synthesis
Ehlers Danlos syndrome

Hereditary Hemorrhagic Telangiectasia



Vessels

IMMUNOGLOBULIN-A
URINE DIPSTICK
HYPERTENSION
IgA NEPHRITIS
HENOCH-SCHONLEIN
PURPURA
PURPURA
ARTHRALGIA
IgA VASCULITIS


A
B
C
D

Platelets

Platelets

Platelet count (per µl)	Diagnosis	Effect
150,000-450,000	Normal	Normal hemostasis
100,000-150,000	Mild thrombocytopenia	Potentially increased oral bleeding from invasive procedures, rare
50,000-100,000	Mod. thrombocytopenia	Possibly increased oral bleeding from invasive procedures, uncommon
25,000-50,000	Severe thrombocytopenia	Problematic bleeding (even with venipuncture) Elective invasive procedures should be deferred
< 25,000	Life threatening thrombocytopenia	Spontaneous bleeding possible Only emergency procedures with hematologist involvement Platelet transfusion

Platelets

QUANTITAVE THOROMBOCYTOPENIAS : Low production of platelets

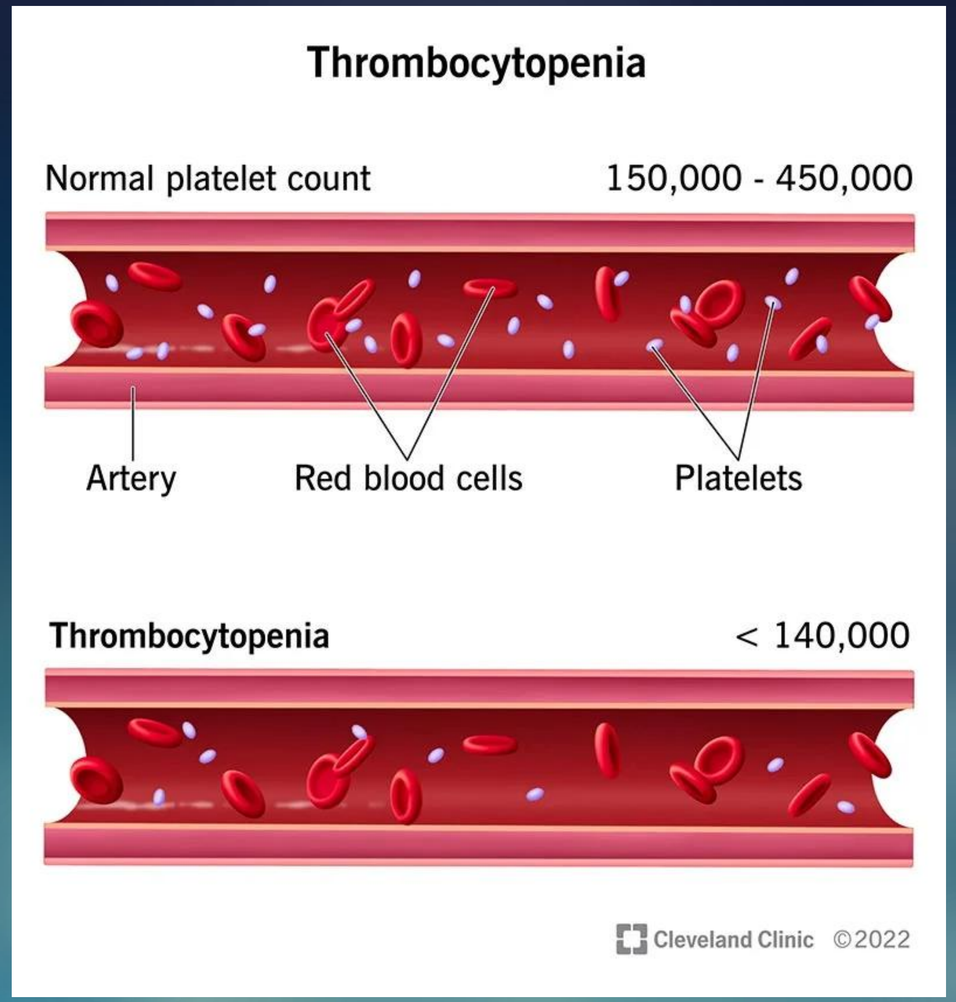
Reduced number of normally functioning platelets (150,000-450,000/µL)

Bone marrow disorders
- Aplastic anemia
- Drug induced/Chemotherapy
- Myeloproliferative syndromes
- Malignancy

Congenital abnormalities (Fanconi's anemia, Alport syndrome, Wiskot Aldrich syndrome)

Nutritional etiologies. (Vit. B12 / follate deficiency, Alcohol abuse)

Other disease states: Sepsis, Liver failure



Platelets

QUANTITAVE THOROMBOCYTOPENIAS: Elevated destruction of platelets

Idiopathic Thrombocytopenic Purpura
Thrombotic Thrombocytopenic Purpura
Disseminated Intravascular Coagulation
Systemic Lupus Erythematosus
HIV Related Thrombocytopenia

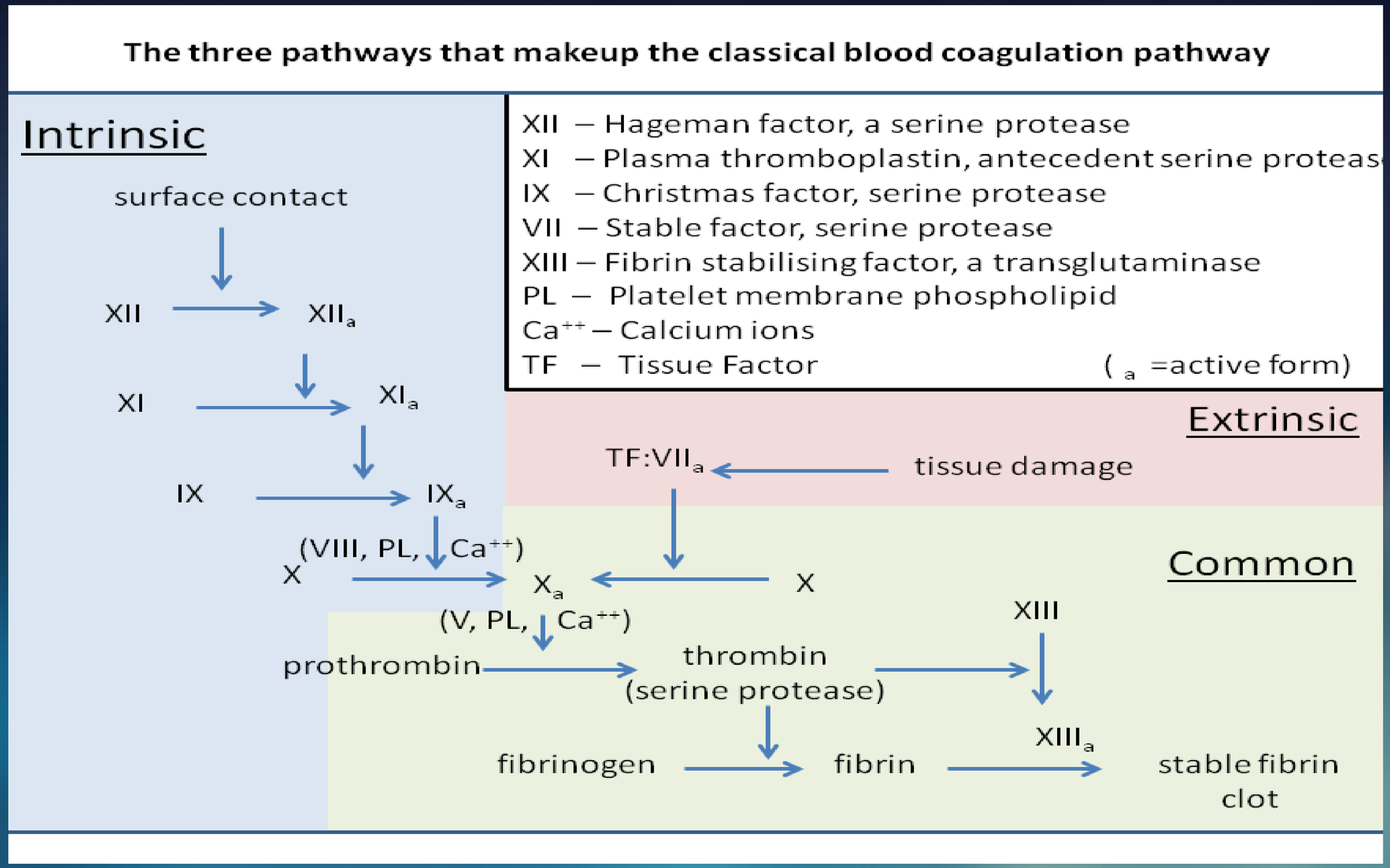
QUALITATIVE THOROMBASTHENIAS: Abnormal adhesion or aggregation

Glanzmann's Thrombasthenia
Von Willebrand disease
Gray platelet syndrome
Scott's syndrome
Platelet inhibition by ASA/Cox inhibitors

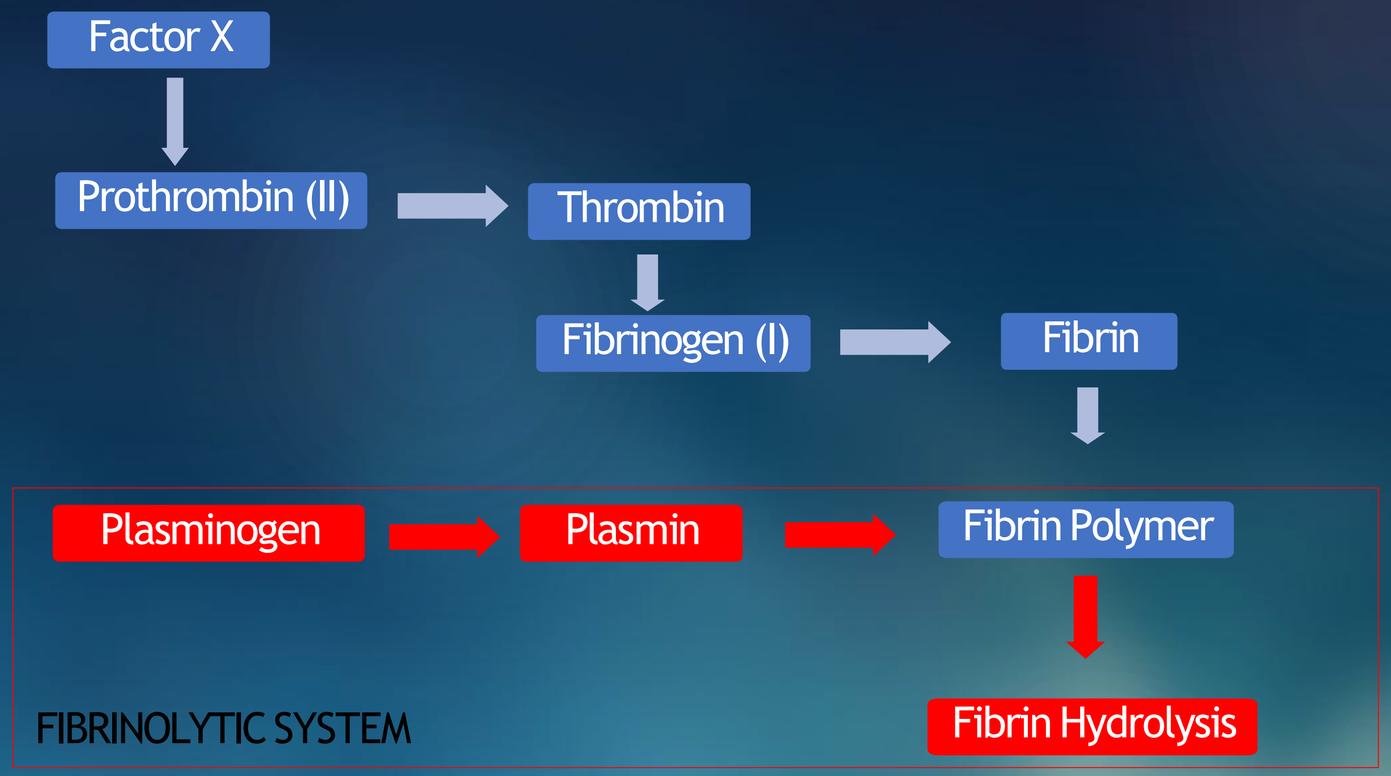
Coagulation Cascade

PTT

PT/INR


The three pathways that makeup the classical blood coagulation pathway
Intrinsic
surface contact
XII
XIIa
XI
XIa
IX
IXa
(VIII, PL, Ca++)
X
Xa
(V, PL, Ca++)
prothrombin
thrombin
(serine protease)
fibrinogen
fibrin
XIII
XIIIa
stable fibrin
clot
XII – Hageman factor, a serine protease
XI – Plasma thromboplastin, antecedent serine proteas
IX – Christmas factor, serine protease
VII – Stable factor, serine protease
XIII – Fibrin stabilising factor, a transglutaminase
PL – Platelet membrane phospholipid
Ca++ – Calcium ions
TF – Tissue Factor
(a =active form)
Extrinsic
TF:VIIa
tissue damage
Common
X

Coagulation Disorders of the Common Pathway



Coagulation Disorders: Congenital

Coagulation Disorders Congenital – Hemophilia A

Factor VIII deficiency

Recessive X-linked, affects mostly men
50-60% of patients have < 1% of normal factor VIII levels in their plasma
Rest of patients have between 1-20% of normal factor VIII in their plasma

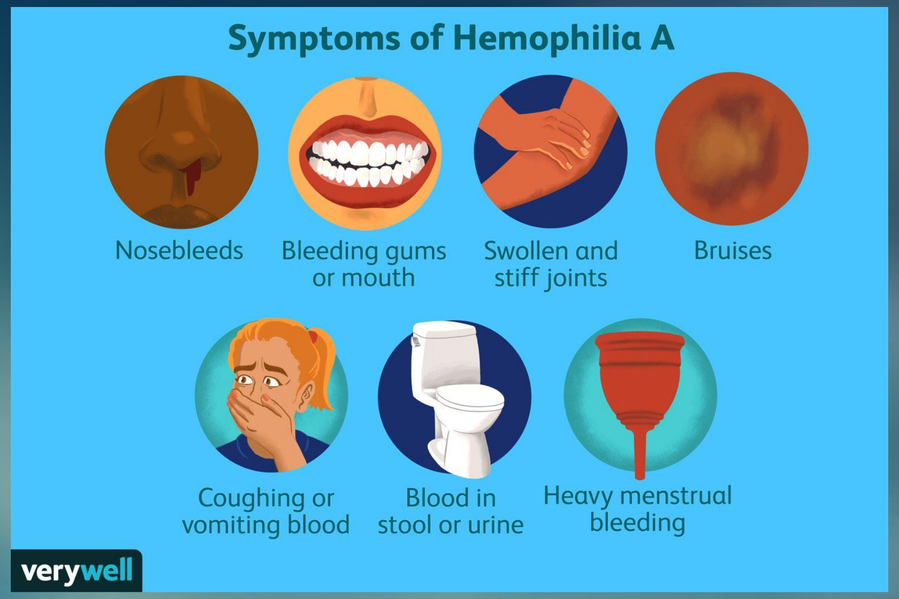
Severe hemophilia patients prior to any surgery require

- factor VIII recombinant replacement 50 U/kg every 8-12h for 7-14 days

Moderate hemophilia patients require

- DDAVP (IV, SC, IN)

All patients require adjunctive antifibrinolytic therapy (Tranexamic acid, ε-aminocaproic acid)
All above dosages need to be adapted according to clinical response and under hematologists guidance gdd



Coagulation Disorders Congenital – Hemophilia B Christmas Disease

Factor IX deficiency

Recessive X-linked coagulation disorder

Affects mostly men

Clinically:

- Severe form (factor IX levels < 1% of normal)
- Moderate form (factor IX levels between 1-5% of normal)
- Mild form (factor IX levels between 6-30% of normal)

All patients require factor IX replacement (dosage depends on factor deficiency)

All patients require adjunctive antifibrinolytic therapy (Tranexamic acid, ε-aminocaproic acid)

Factor dosages need to be adapted according to clinical response

PHARMA

Sporting a $3.5M price tag, CSL and uniQure's hemophilia B gene therapy crosses FDA finish line

By Zoey Becker • Nov 23, 2022 10:42am

Coagulation Disorders Congenital – Hemophilia C Rosenthal Sydrome

Factor XI deficiency

Autosomal (varied penetration) genetic disorder characterized by low levels of factor XI

Affects both sexes

Clinically mild disease characterized by frequent nose bleeds, prolonged bleeding after injury, heavy menstruation

Intervention prior and after surgery involves transfusion of fresh frozen plasma or recombinant factor XI

All patients require adjunctive antifibrinolytic therapy (Tranexamic acid, ε-aminocaproic acid)

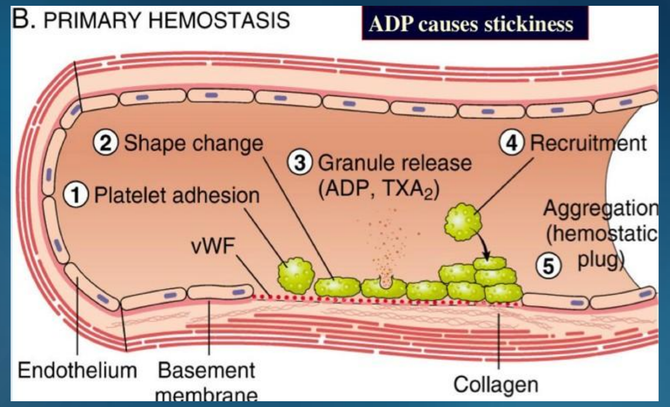
Coagulation Disorders Congenital – Von Willebrandt Disease

A bleeding disorder due to absence or defective vWF

vWF is produced in endothelial cells and megakaryocytes

It is found in circulating plasma, platelets, endothelial cells and sub endothelial connective tissue It induces the adhesion of platelets to the injured vascular wall

It binds factor VIII keeping it in the plasma otherwise factor VIII plasma concentration diminishes



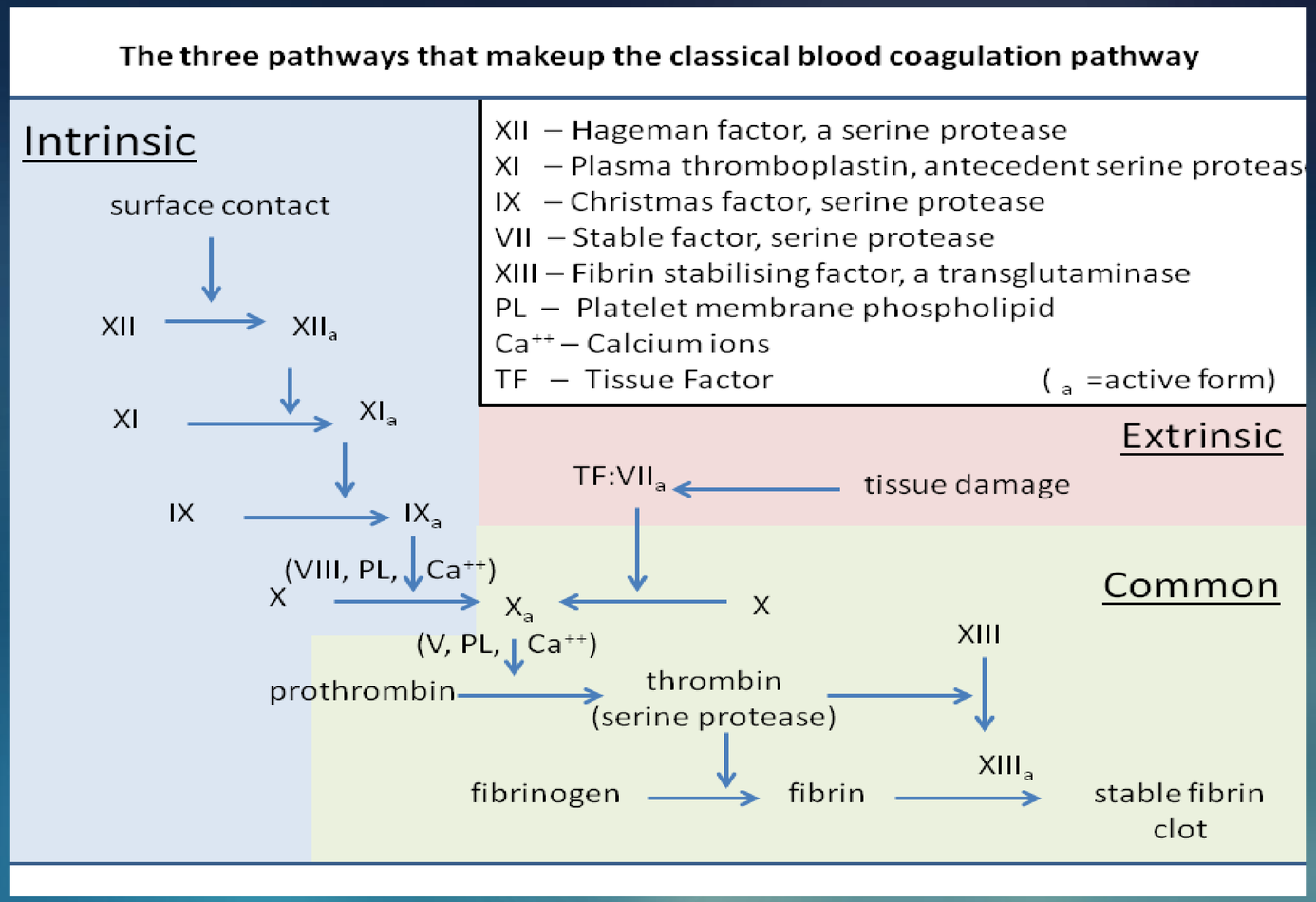
VWD Type 1: DDAVP (IV, SC, IN)
DDAVP stimulates the endogenous release of VWF from endothelial cells, and as a result, also increases factor 8 levels

VWD Type 2 and 3: VWF concentrate

Coagulation Disorders Congenital – Others

Other factor deficiencies

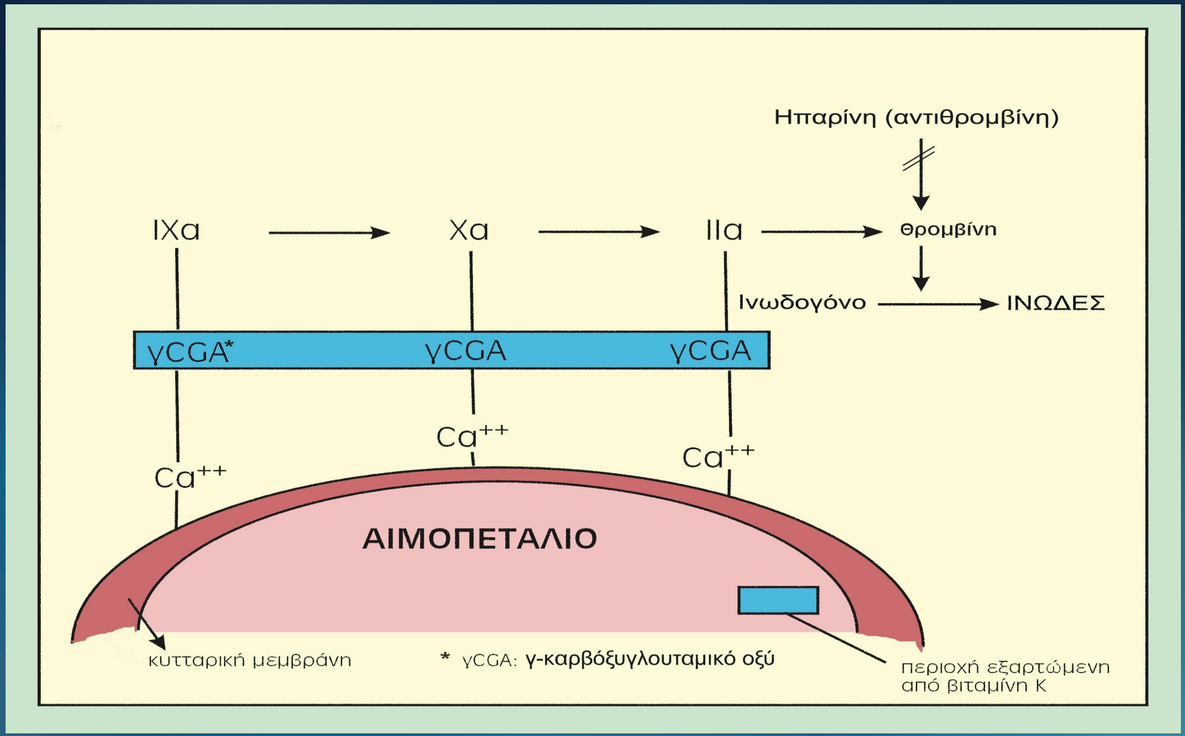
Factor V deficiency
Factor X deficiency
Factor XII deficiency



Coagulation Disorders: Acquired

Coumarins, Warfarins - Vitamin K

Lack of vitamin K leads to lack of γ Carboxy Glutamic Acid making the activation of IX, X and II impossible

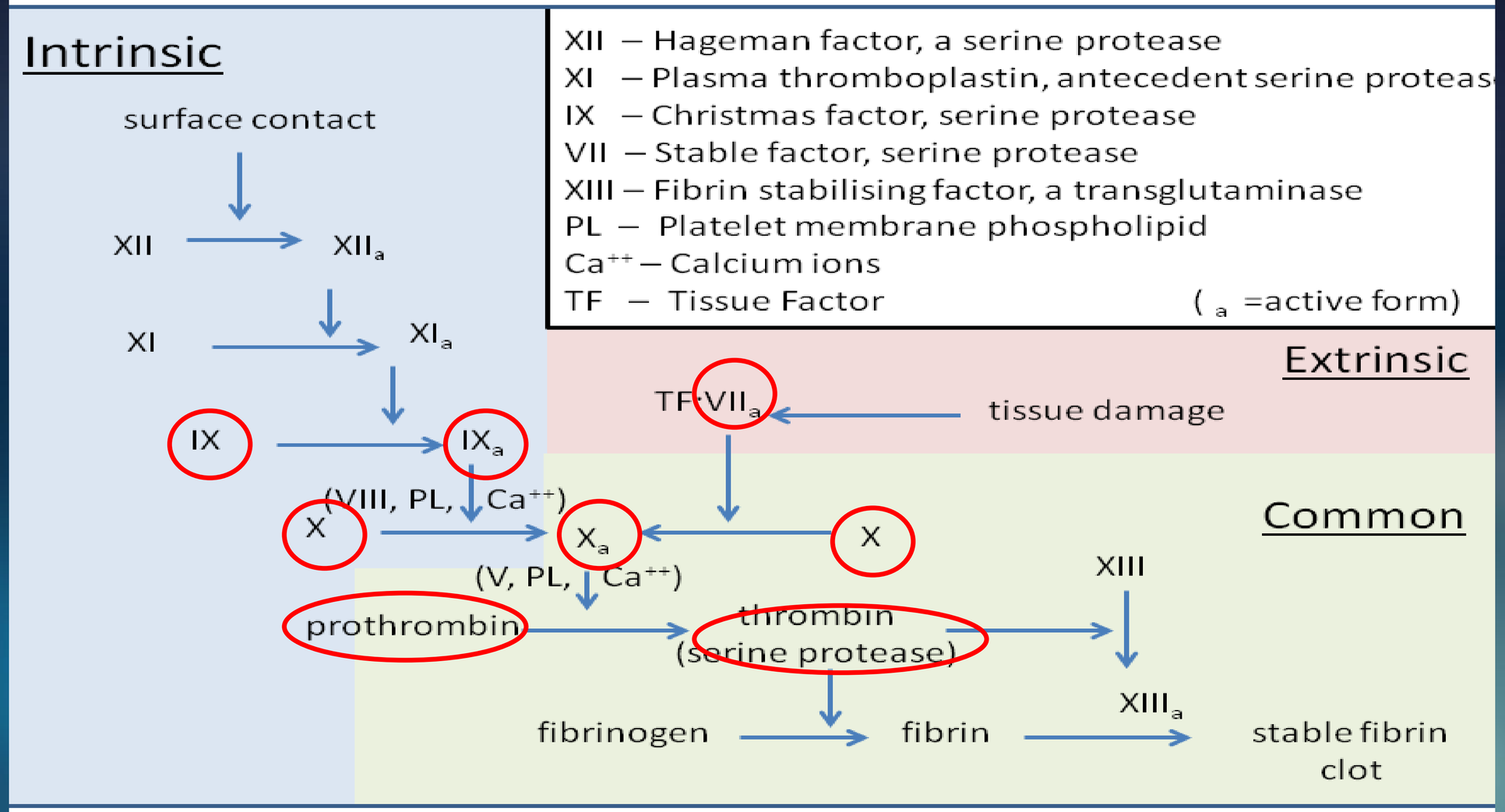


Vitamin K is necessary for

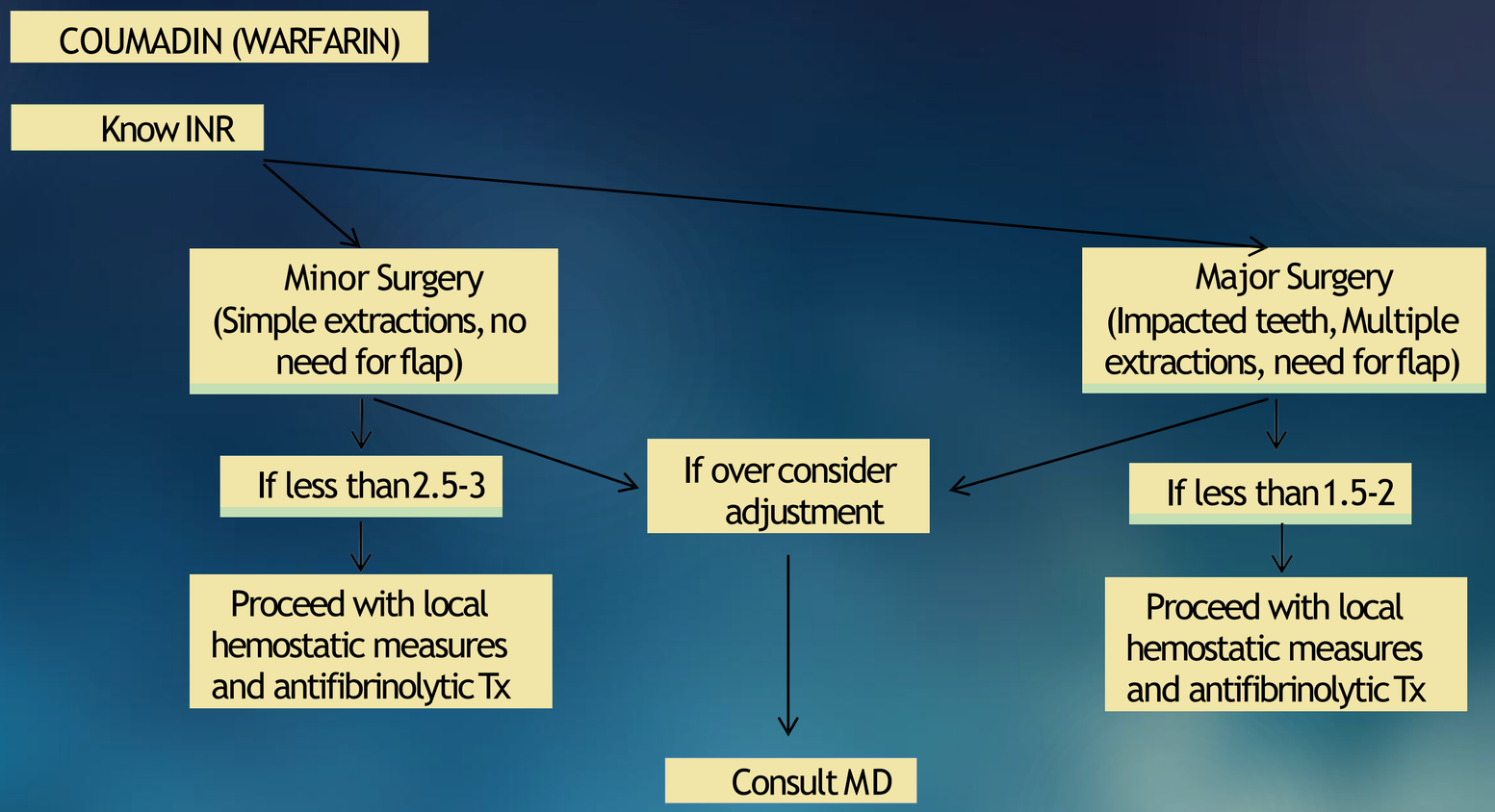
2, 7, 9, 10 and Protein C and S

Coumarins
Warfarins

The three pathways that makeup the classical blood coagulation pathway



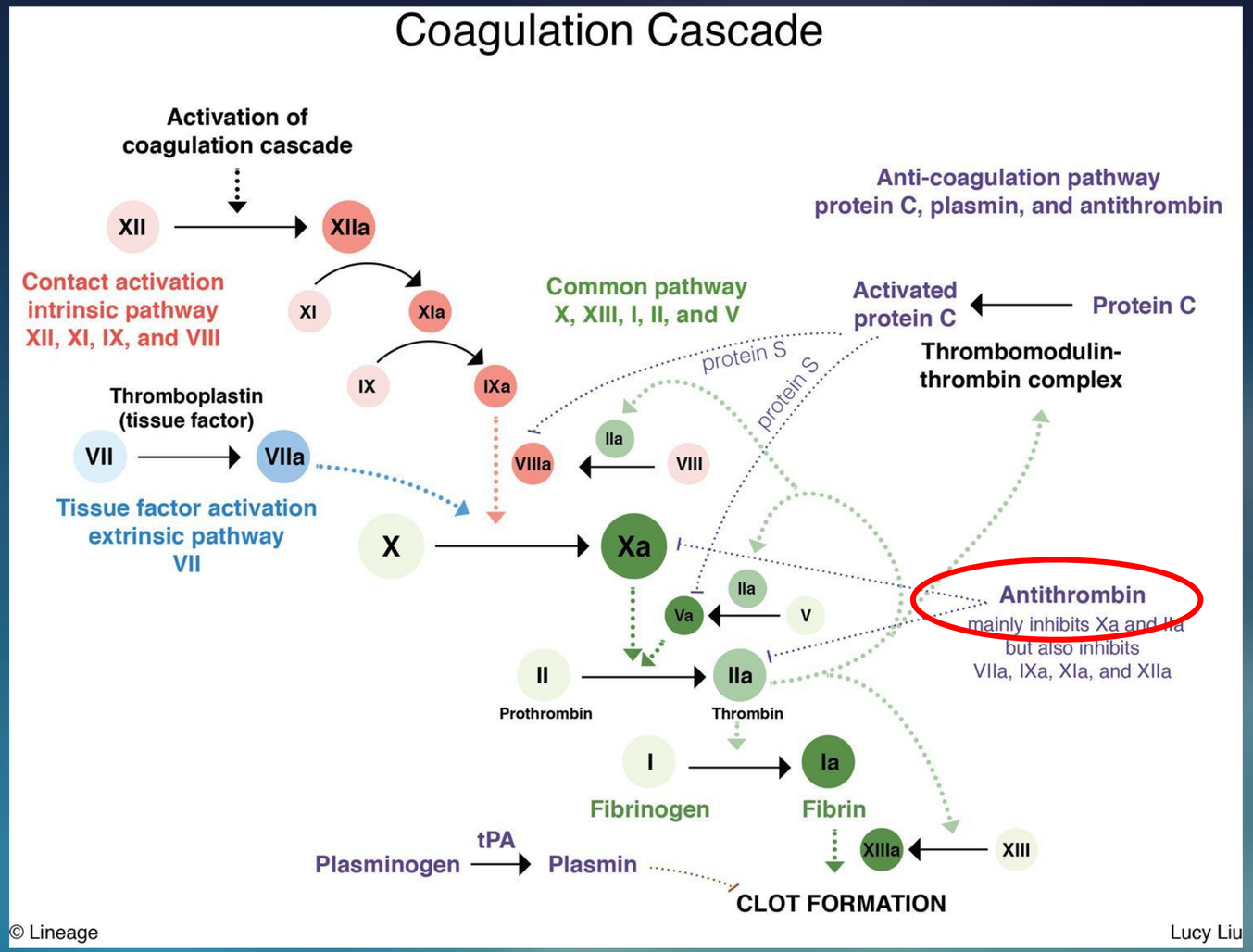
Coumarins, Warfarins - Vitamin K



Heparins

Heparin activates antithrombin both by inducing conformational changes in the protein that specifically enhances factor Xa binding and by providing a surface to promote thrombin or factor Xa binding alongside antithrombin in a ternary bridging complex

Protamine sulfate for reversal of UFH and LMWH

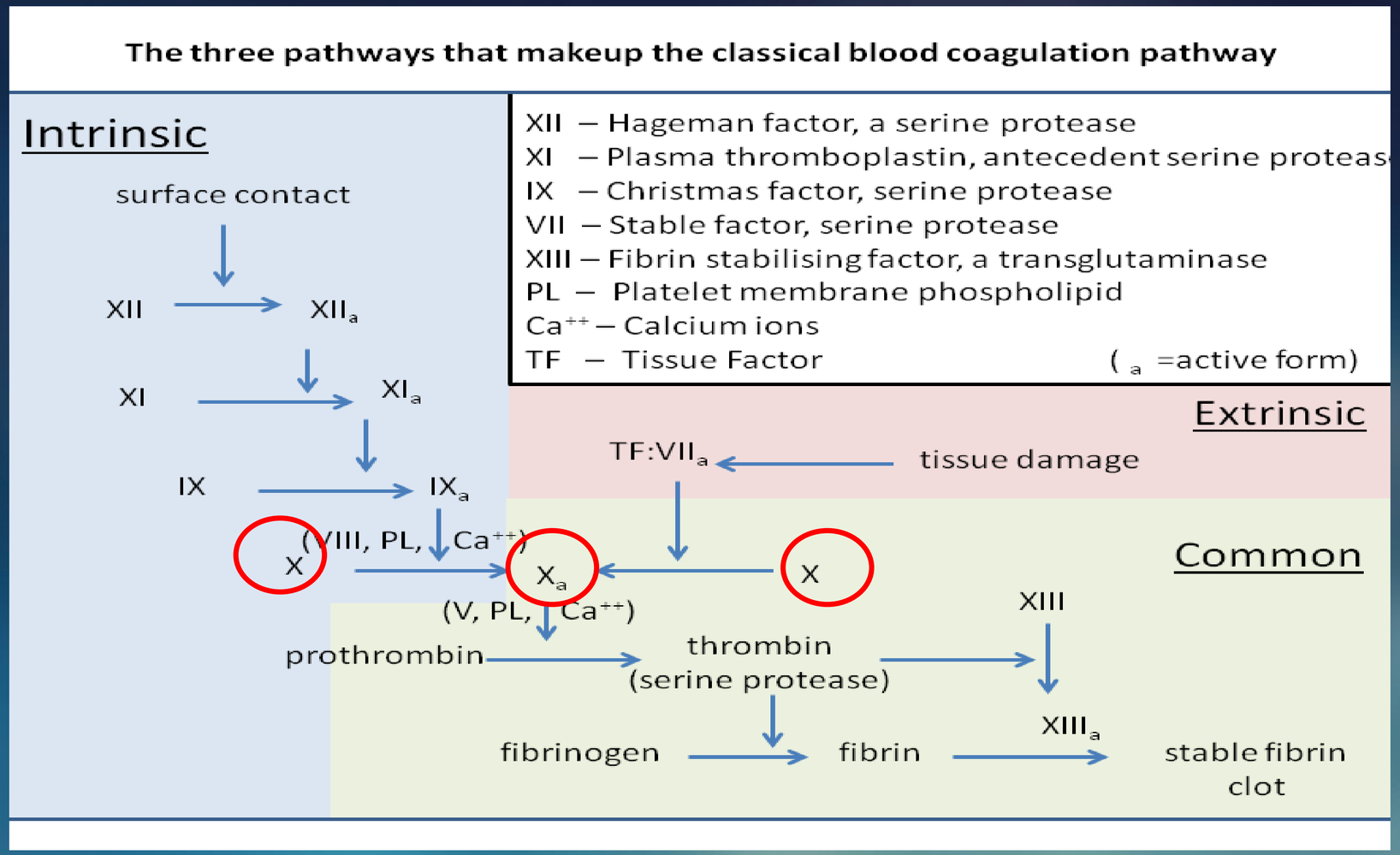


Direct Factor X Inhibitor

Rivaroxaban (xarelto)
Apixaban (eliquis)
Edoxaban (lixiana)

Hepatic metabolism

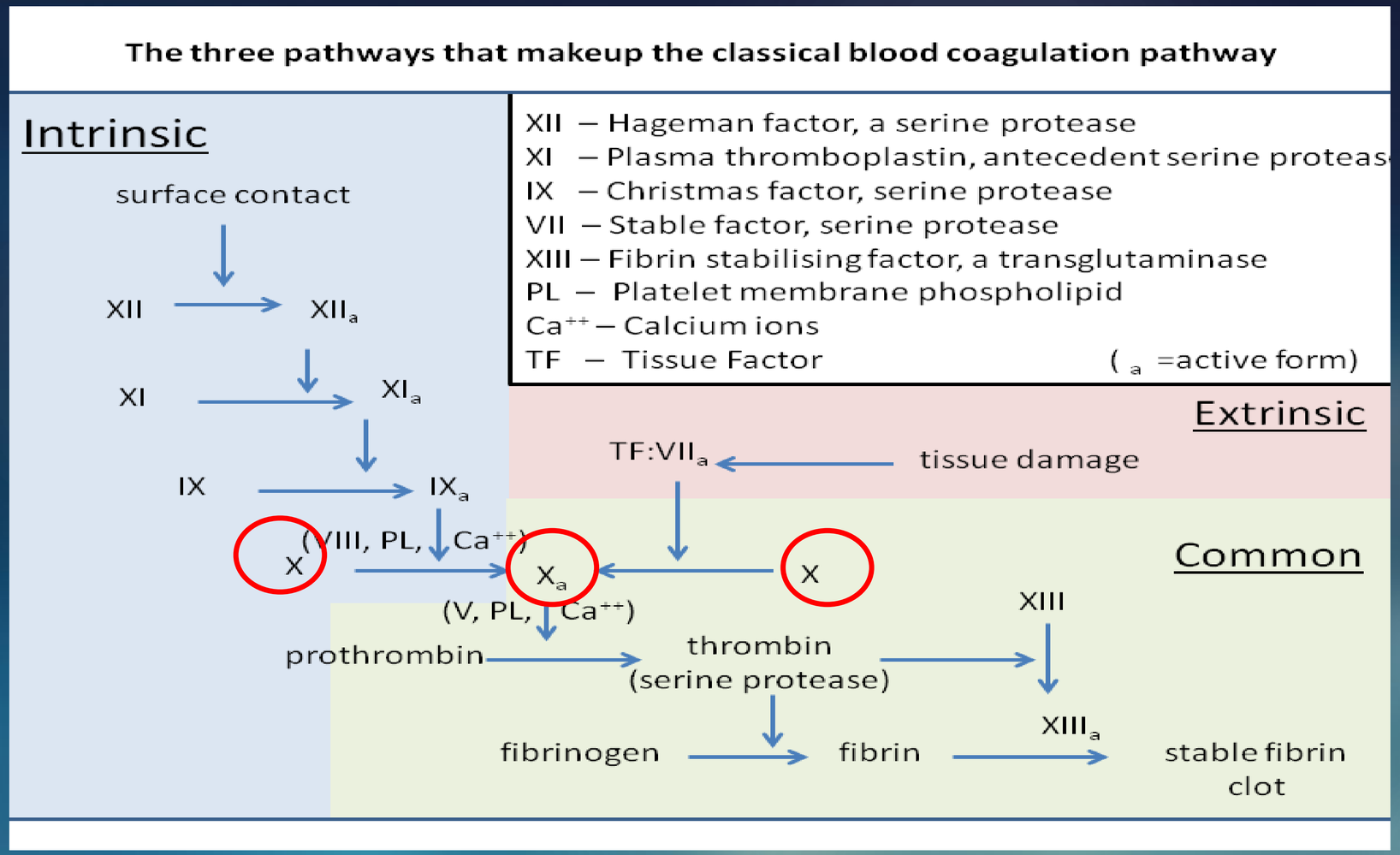
Direct oral anticoagulants (DOACs) are oral medications that specifically inhibit factors IIa or Xa. They are also known as new oral anticoagulants (NOACs) or target-specific oral anticoagulants (TSOACs)



Direct Factor II Inhibitor - Direct Thrombin Inhibitors

Dabigatran (pradaxa)
Argatroban
Inogatran
Melagatran
Dabigatran

Renal Metabolism



Reversal of DOACs

Only patients with life-threatening bleeding, and major uncontrolled bleeding with hemodynamic compromise are candidates for the use of reversal agents for DOACs

- Idarucizumab to reverse dabigatran
- Andexanet alfa to reverse apixaban and rivaroxiban

Non-major bleeding: using local hemostatic measures, supportive care, withholding next dose of DOAC, awaiting spontaneous clearance are usually sufficient to control the bleed

HEPARIN+PTT (10 LETTERS)
With H also Hemophilia

COUMADIN+PT (10 LETTERS)
WARFARIN +PT
VITAMINK +PT
LIVERBAD +PT

Antithombotic treatment

Agent	Action	Indications	Administration	Monitoring	Reversal	Time to stop prior to Sx
Coumadin	Vitamin K antagonist. Inhibition of IX, X, II, VII	Prosthetic valves DVT, A Fib. Pulmonary embolism	Oral	INR	• Vitamin K • Fresh frozen plasma • Prothrombin complex concentrate	Generally 5 days depending on INR value
Heparin Unfractionated	Antithrombin	As above	IV	INR	Protamine Sulfate	Approx. 2 hours
LMWH	Antithrombin	As above except	SC	INR	Protamine Sulfate	One day
Dabigatran (Pradaxa)	Antithrombin	Afib, Post hip, Knee Sx DVT prevention	Oral	Not needed	Available	24 Hours ??
Apixaban (Eliquis)	Factor X Inhibitor	Afib, Orthop Sx DVT prevention	Oral	Not needed	Available	24 Hours ??
Rivaroxaban (Xarelto)	Factor X inhibitor	Afib, Orthop Sx DVT prevention	Oral	Not needed	Available	24 hours??

Antiplatelet Treatment

Agent	Action	Indications	Administration	Monitoring	Reversal	Time to stop prior to Sx
Aspirin (ASA)	Cox inhibitor Prevents thromboxane production Irreversible platelet inhibition	MI prevention	Oral	Bleeding time	Not available	Depending on dose. No stopping low dose ,minor surgery High dose, extensive surgery stop 5-7 days
NSAID's	Cox inhibitors Prevents thromboxane production Reversible platelet inhibition	Anti-inflammatory drugs Pain Management	Oral, Rectal IV	Bleeding time	Not available	1-2 days for extensive procedures
Clopidogrel (PLAVIX)	ADP platelet inhibition No aggregation	Prevention of MI Post stent tx.	Oral/IV	Bleeding time	Not Available	5 days (esp. for elective Surgery)
Ticlopidine (Ticlid)	As in Plavix	As in Plavix	Oral	Bleeding time	Not available	10-14 days (esp. for elective Surgery)

Fibrinolytics

Agent	Action	Indications	Administration	Monitoring	Reversal	Time to stop prior to Sx
Streptokinase Urokinase	Potentiates Plasminogen activation	MI Tx DVT Tx Pulmonary embolism	IV	Symptom improvement	Not available	Consult MD
R-tPAs Alteplase Retelplase	Potentiates Plasminogen activation	MI Pulmonary embolism Tx Ischemic Stroke	IV	Symptom improvement	Not available	Consult MD

Hemostatic Agents

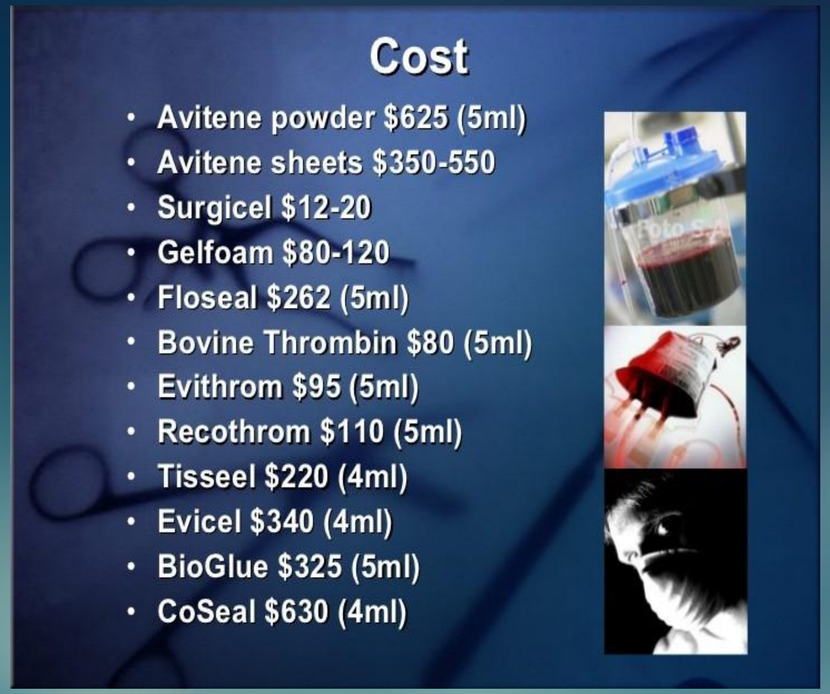
Local Hemostatic Agents

- Help in achieving hemostasis
- Adjuncts in managing anticogulated patients
- Allow surgical treatment without complete reversal
- Minimize blood loss
- Avoid blood transfusions

Local Hemostatic Agents

Selection considerations

- Material properties (powder, fabric, sponge, paste, liquid) (2 stage pastes/liquids)
- Origin (Human, Bovine, Porcine, Plant, Recombinant)
- Ease of use (delivery to site)
- Speed of action
- Effectiveness of hemostasis
- Origin (Human, Bovine, Porcine, Plant, Recombinant)
- Immunogenicity/Allergenicity
- Absorption rate
- Cost, availability, storage



Local Hemostatic Agents

Collagen products (Avitene, Collaplug)

Bovine origin collagen shredded in fibrils

Promotes platelet aggregation-degranulation

Potentially immunogenic

Local Hemostatic Agents

Gelatin products (Gelfoam, Surgifoam)

Solid:

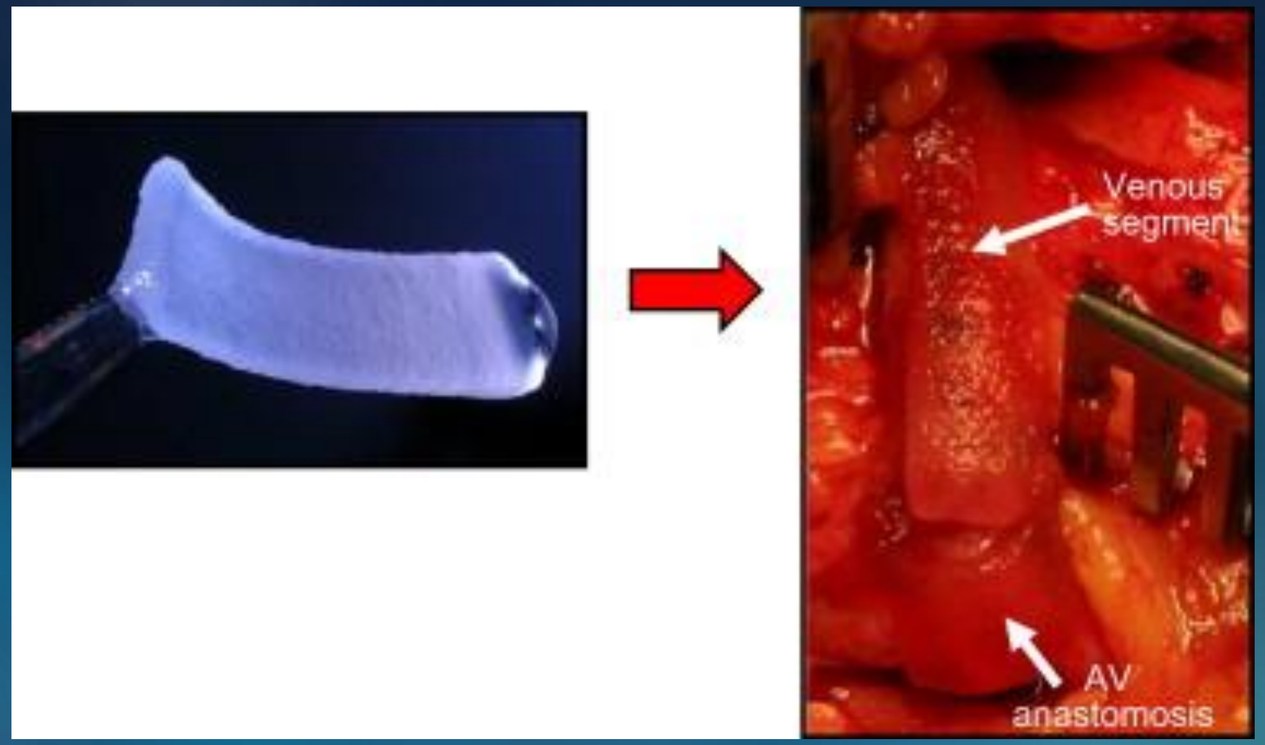
- Porcine origin water insoluble gelatin
- Provides the matrix for platelet adhesion-aggregation
- Promotes the formation of platelet plug
- “Foamy” consistency
- Absorbs blood (45 times it’s weight)
- Difficult to handle
- Liquefied within a week-absorbed in 4-6 weeks
- Very difficult to handle once wet

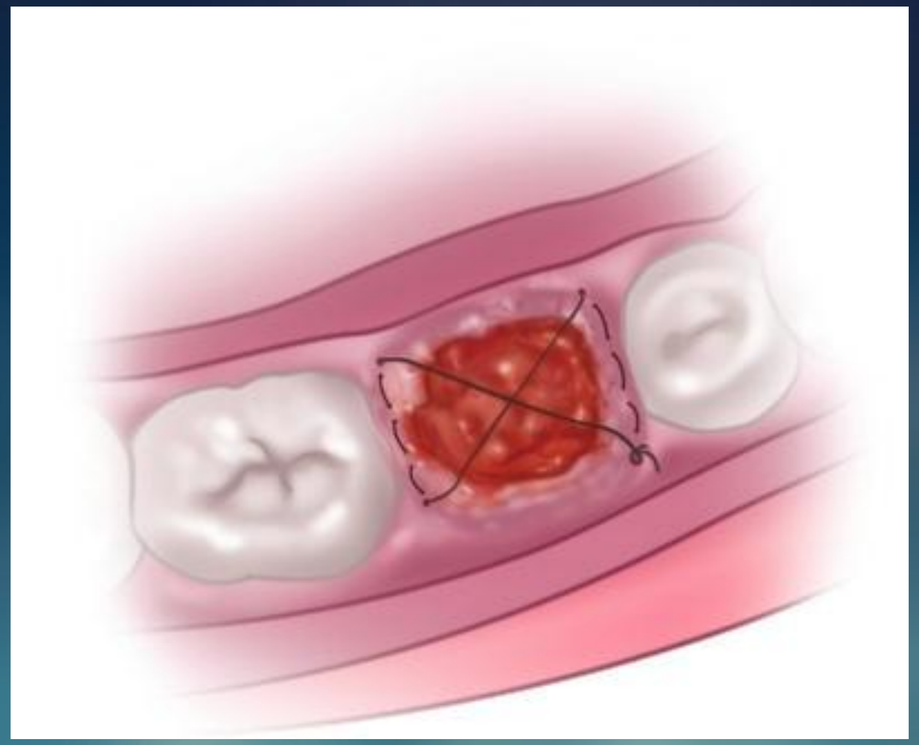
Flowable Gelatins (Floseal, Surgiflo)

- Same function
- Flow and cover wide areas of bleeding

Local Hemostatic Agents

Local Hemostatic Agents





Local Hemostatic Agents

Oxidized Regenerated Cellulose (Surgicel)

Plant origin-non immunogenic

Cloth like format

Easy to handle

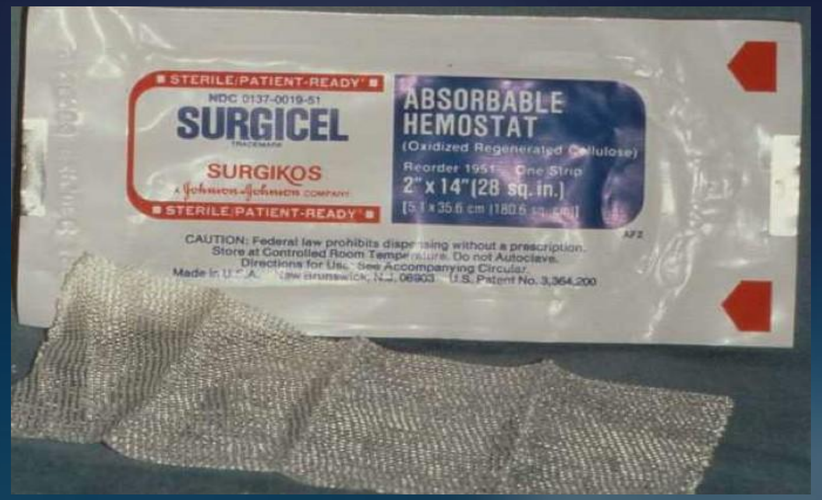
Absorbs blood and becomes a low pH gel causing local vasoconstriction

Provides matrix for platelet adhesion and formation of platelet plug

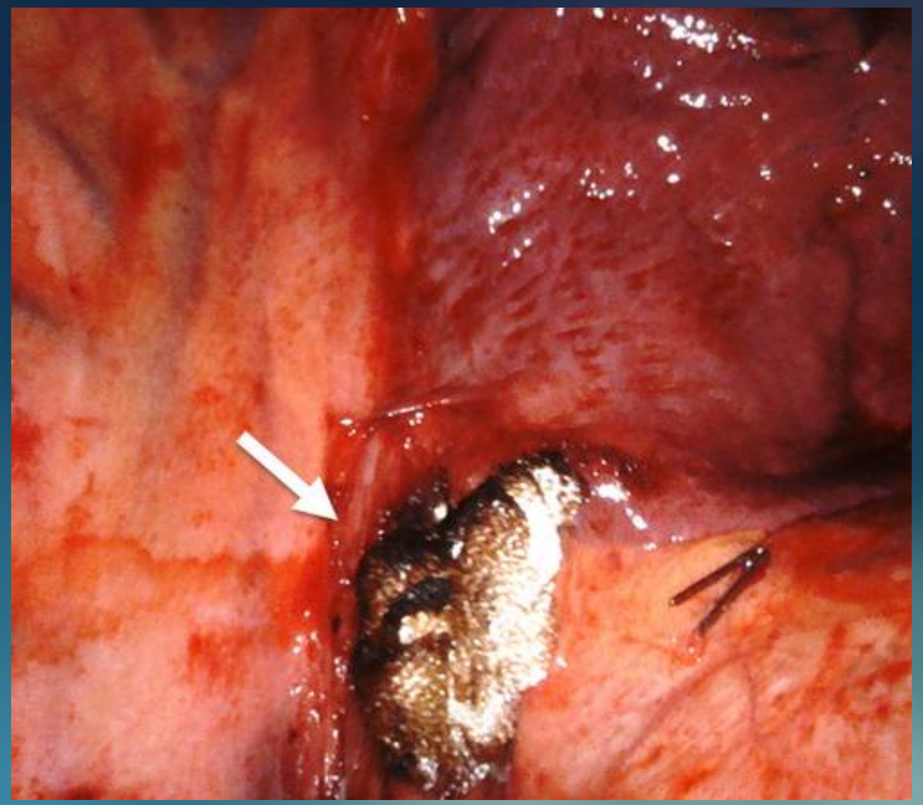
Acts within 6-8 minutes

Low pH is bactericidal but interfere with bone healing and nerve damage

Absorbable in 7-14 days



Local Hemostatic Agents



Local Hemostatic Agents

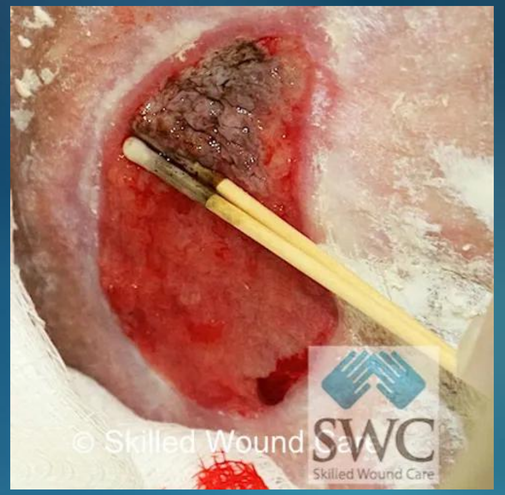
Styptics (Silver Niterate Sticks)

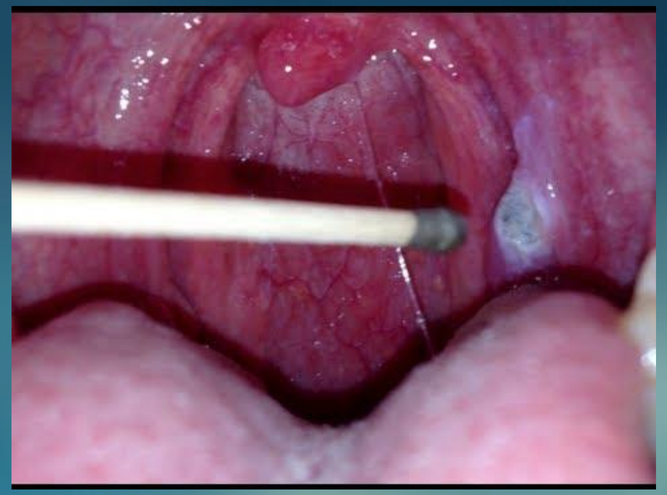
Chemical burn locally

Tissue protein denaturation

Vessel occlusion

Potential for tissue damage



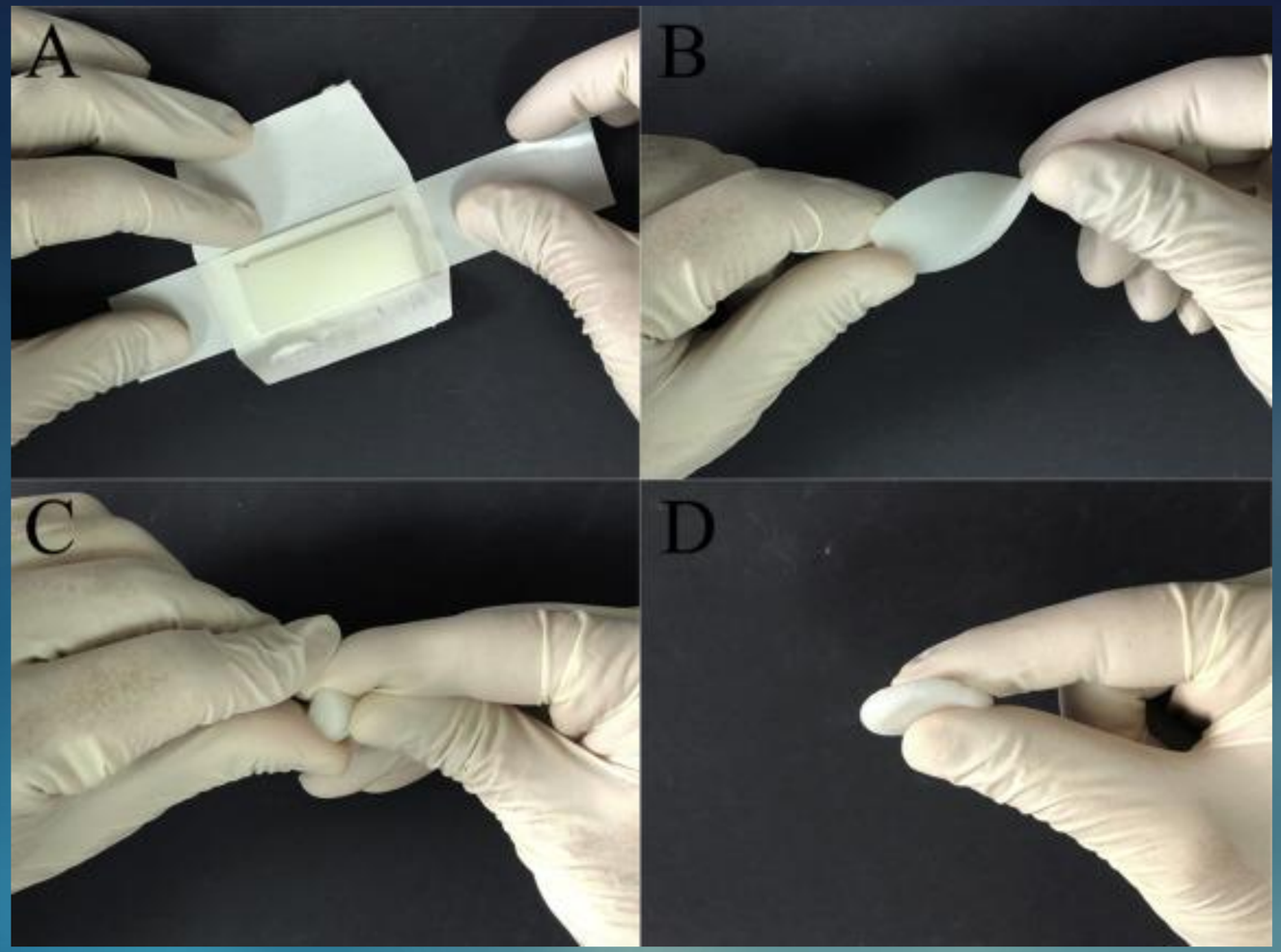


Local Hemostatic Agents

Bone Wax

For bleeding in bone only
Mechanical occlusion of bleeding vessels
Sterile bees wax

Foreign body in socket
- Delays healing
- Nidus of infection
- Foreign body (giant cell) reaction



Local Hemostatic Agents

Thrombin preparations

Converts fibrinogen into fibrin

Bovine, Human or recombinant preparation

Potentially immunogenic (bovine origin)

Allergic reaction second time around

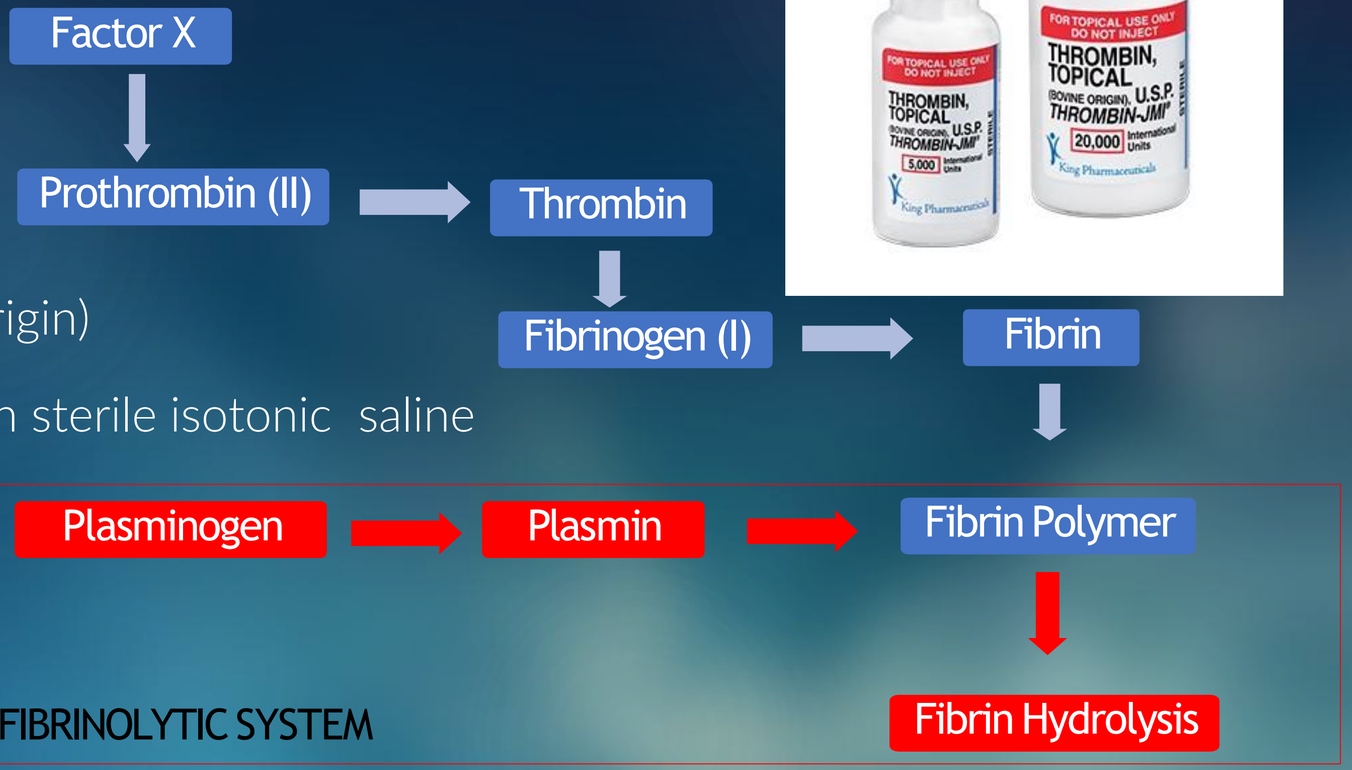
Potential risk for viral transmission (human origin)

Lyophilized powder needs reconstitution with sterile isotonic saline

Fast action

Liquid preparations need refrigeration

Cost

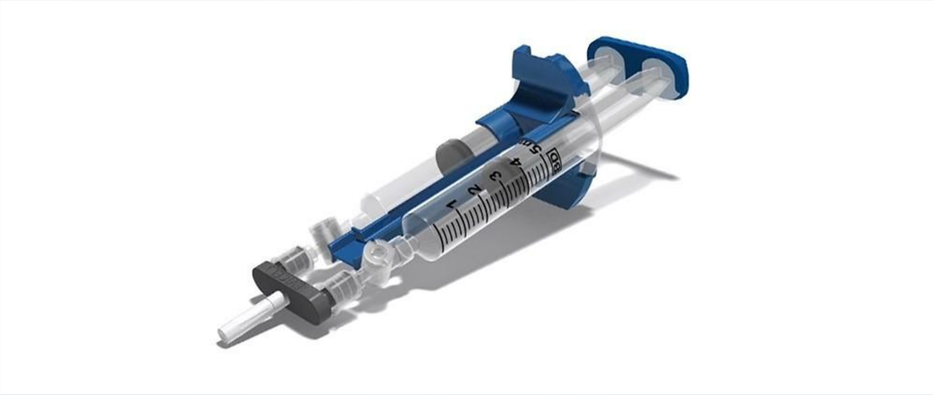


Local Hemostatic Agents

Fibrin Sealants (Tisseel, Evicel)

Two syringe system, self mixing, one containing thrombin and the other fibrinogen
Available both, of human and bovine origin

Common Pathway



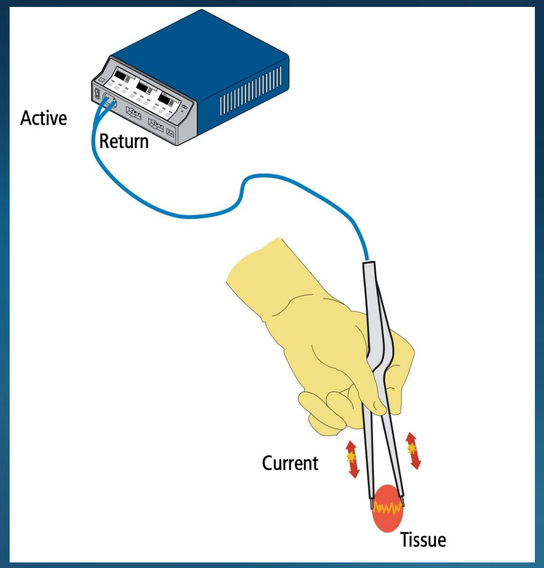
Local Hemostatic Agents

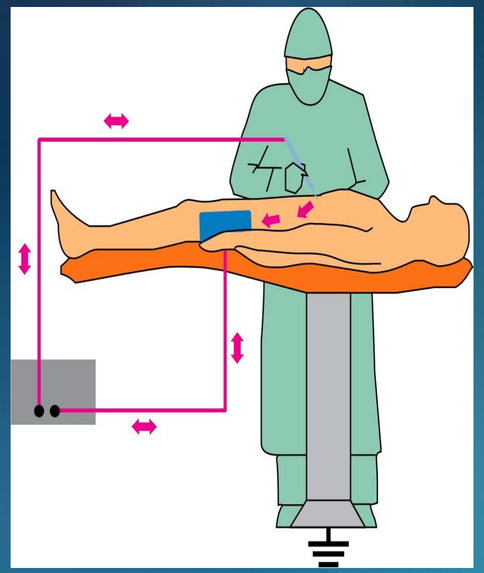
Thermal methods

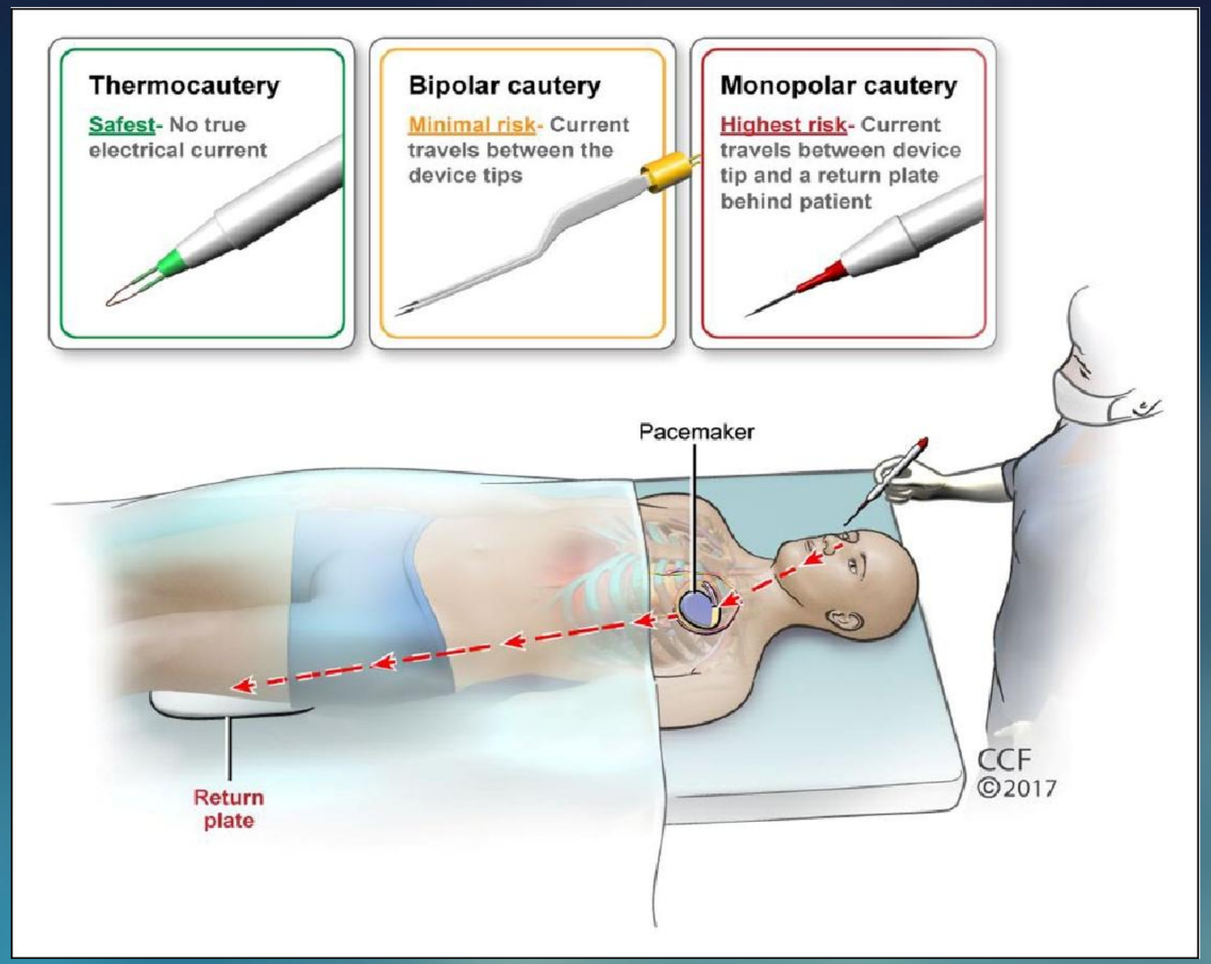
-Electrocautery

monopolar

bipolar







Hemostatic- Antifibrinolytics

Inhibition of the pathway Augments Coagulation

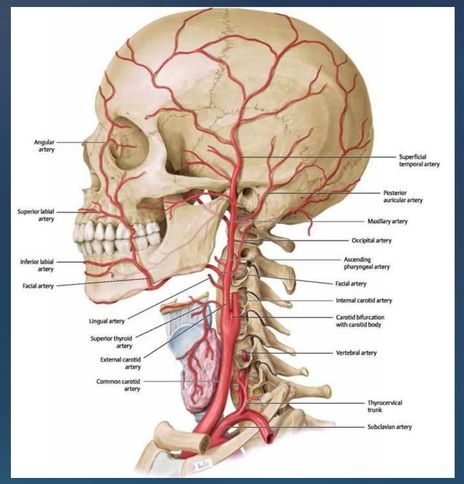
Aminocaproic Acid – Plasmin Inhibitor
systemically (oral or IV) 75mgr/kg (Up to 6 gr) qid X 7-10 days
oral rinse (1.25g/5ml). rinse with 5 ml for 30 sec then repeat qid X 5-7days

Tranexamic Acid – Plasminogen Inhibitor
systemically, oral (25mgr/kg tid/qid) or IV (10mgr/kg 3-4 times a day X 2-8 days oral rinse (4.8% solution), rinse with 10ml for 2 min qid X 5-7 days

Effective adjunct hemostatic therapy to prevent or treat excessive bleeding in patients with bleeding disorders, surgical procedures with significant blood loss (cardiac surgery, orthopedic surgery).

Postoperative Complication

Bleeding

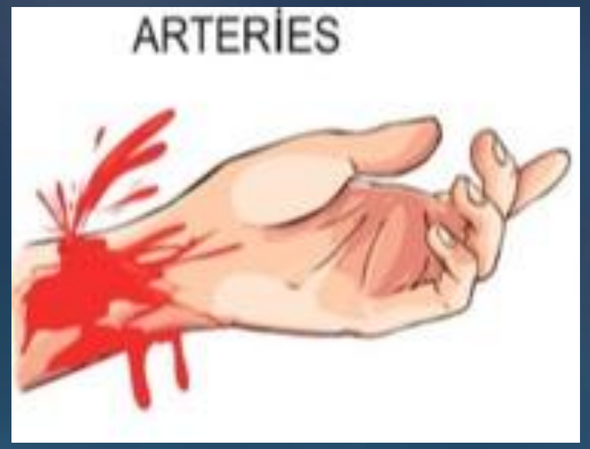


Vascular
-arteries
-veins
-capillaries

Soft Tissue
-inflamed tissue
-mucosal tears

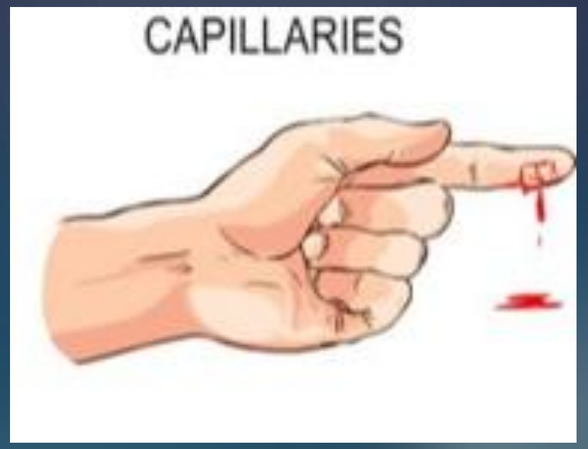
Bone
-nutrient canal
-central vessels (ex IAA)
-marrow
-central vascular
lesions (Hemangiomas, AVMs)

Bleeding



Arterial bleed
-pulsatile
-brisk
-bright red

Venous bleed
-even stream
-dark blood



Capillary bleed
-slow ooze
-bleeding point hard to make out
-bright red

Bleeding

Postoperative Complications

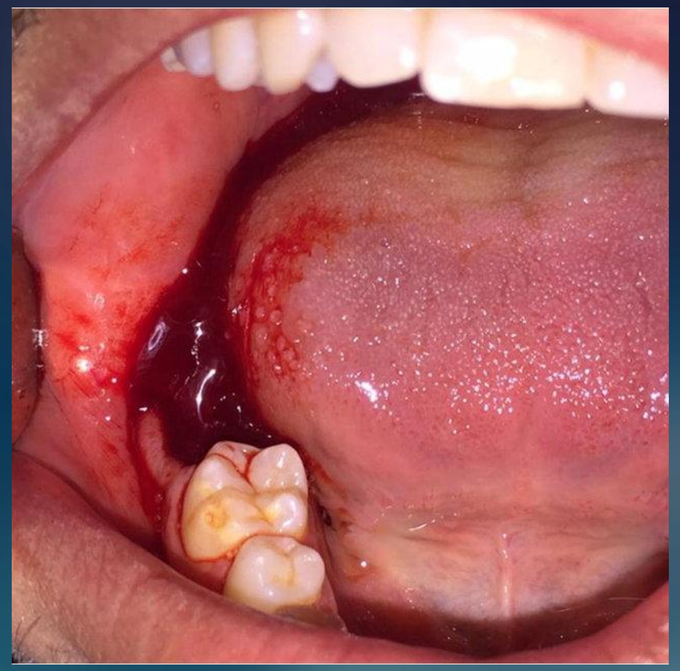
Post op bleeding can be reactionary or secondary

<u>Reactionary (within first 24 hours post op)</u>

Clot formation 6-12 hours post-op
Oozing usually resolves within 36-72 hours and should respond to pressure

Occurs as a result of mechanical dislodgement of the hemostatic plug in the vast majority of cases:

- Overzealous rinsing
- Trauma
- Spitting
- Smoking
- Chewing hard food
- Vomiting
- Increased intraluminal pressure
- Vasoconstrictor degradation
- Left over granulation tissue
- Anticoagulation medication



In a few patients it might be due to unrecognized before coagulation disorder, or systemic disease

Bleeding

Postoperative Complications

Post op bleeding can be:

<u>Secondary (develops after the first 24 hours)</u>

This occurs as a result of failure of the hemostatic plug and the healing process in the vast majority of cases:

- Infection
- Trauma
- Nonspecific fibrinolytic activity

If "liver clot" present careful removal is needed to control bleeding

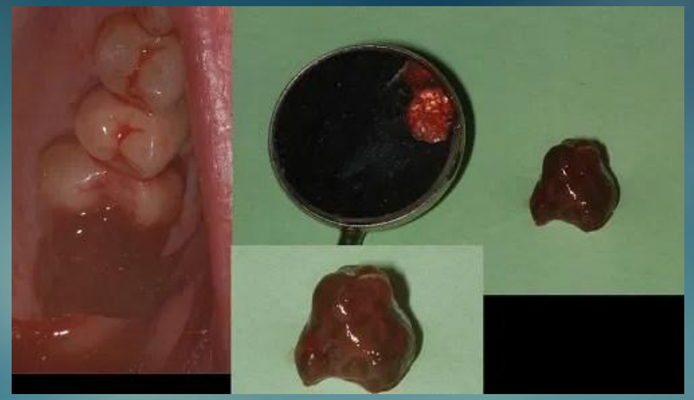
Dark red jelly like mass that forms over tooth socket, does not stop the bleeding from the socket underneath

Treatment:

-Remove by high speed suction or large curette

-irrigate socket

-apply pressure with gauze, local measure prn



Bleeding

Postoperative Complications

Prevent:

Apply pressure

Get an idea as to the amount of blood lost
- Tachycardia
- Blood pressure
- Orthostatic hypotension
- Thirst

Review medical history once more

Determine source
- bone
- soft tissue

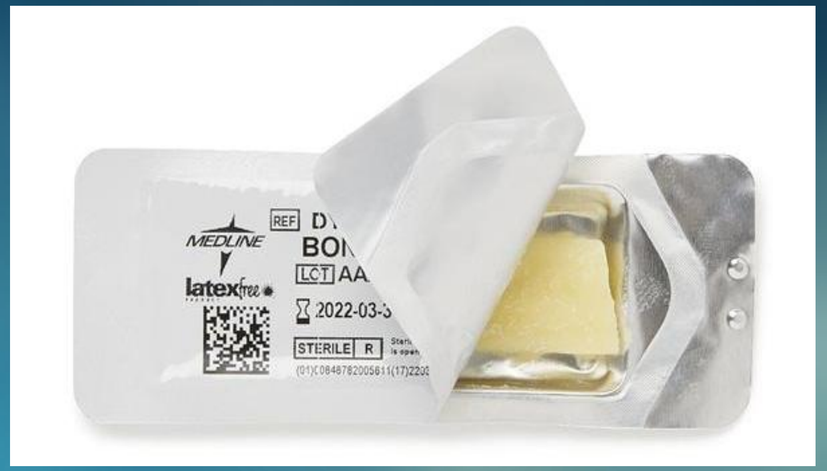
Determine vessel type
- artery
- vein

Bleeding

Postoperative Complications

Bone

Apply Pressure → Crush Bone If Point Source → Bone Wax → Hemostatic Agents



Bleeding

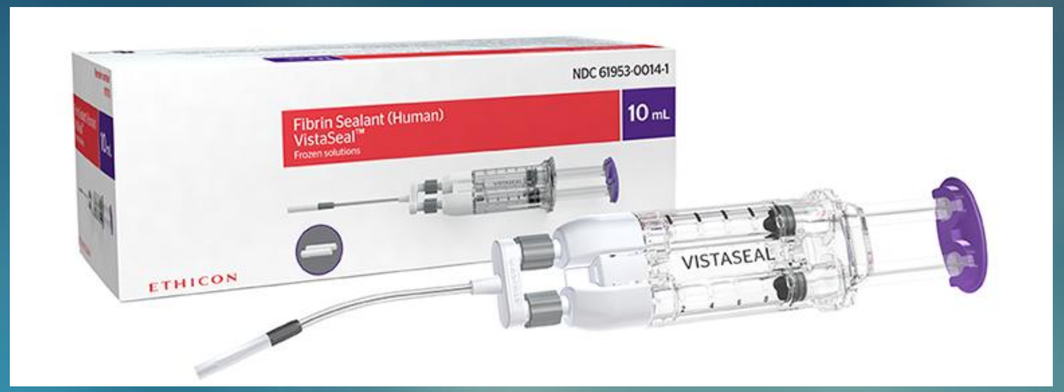
Postoperative Complications

Soft Tissue

Apply Pressure → Identify Vessel → Cauterize / Ligate → Apply warm pack if generalized → Hemostatic Agents → Fribrin Sealant

Prevention

- Avoiding flap tears
- Primary wound closure
- Topical agents: Gelfoam and Surgicel



If the bleeding persists, what do you do?

- Infiltration with local anesthetic (1:100,000 epinephrine)
- Existing clot should be curetted/suctioned from socket to allow for visibility
- Placement of hemostatic agents into socket (Gelfoam, Surgicel, or Avitene) and suture + pressure with gauze
- If hemostasis is still not achieved, get coagulation screen and hematology consult to investigate possible underlying cause

Thank you

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